

BORONGAN, PhilippinesWMO: **985530**Lat: **11.660N**Lon: **125.442E**Elev: **10**StdP: **14.69**Time Zone: **8.00 (E08)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	72.4	73.5	70.1	110.8	74.8	71.2	115.3	75.4	15.4	79.7	14.0	80.3	3.0	270	0.315

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	10.7	91.4	81.9	89.9	81.7	89.0	81.5	82.8	88.7	82.3	88.1	81.9	87.4	6.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.0	161.0	86.9	80.6	159.1	86.7	80.1	156.4	86.0	46.6	89.7	46.1	88.9	45.7	88.1	
(p)															

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.3	11.5	9.8	68.2	95.7	4.1	3.3	65.3	98.1	62.9	100.0	60.6	101.9	57.6	104.3
			67.2	84.1	3.7	0.5	64.5	84.4	62.3	84.7	60.2	85.0	57.5	85.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.1	79.1	79.2	80.1	81.2	82.2	82.2	81.8	82.5	82.0	81.5	80.9	80.4	
	DBStd	1.97	2.05	1.66	1.57	1.34	1.29	1.55	1.73	1.82	1.68	1.45	1.57	1.62	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD50	11349	903	816	932	935	998	966	985	1008	961	977	926	942	
	CDD65	5874	438	396	467	485	533	516	520	543	511	512	476	477	
	CDH74	59705	3707	3363	4406	5131	6005	5690	5503	6008	5376	5285	4732	4499	
	CDH80	17392	632	592	1005	1547	2089	1937	1797	2178	1787	1619	1248	961	
Wind	WSAvg	3.8	4.9	4.9	4.6	4.1	3.4	3.2	3.1	3.1	3.1	3.4	3.7	4.5	
Precipitation	PrecAvg	158.7	23.7	18.0	14.6	8.8	7.9	10.2	9.0	7.0	7.8	11.3	18.3	25.4	
	PrecMax	256.1	46.1	64.0	49.8	20.9	16.4	17.7	16.1	15.4	13.3	19.3	33.9	46.8	
	PrecMin	101.4	6.9	2.6	2.6	2.2	1.8	4.1	3.0	2.2	3.5	6.5	8.3	10.7	
	PrecStd	43.0	10.2	14.3	10.3	5.4	4.3	3.5	3.3	3.3	2.6	3.4	6.9	9.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.3	86.2	87.3	88.8	90.8	91.6	92.3	93.2	92.4	90.8	88.8	87.8	
		MCWB	80.5	80.3	80.8	81.2	81.9	82.1	81.8	82.6	82.4	81.5	81.7	81.0	
	2%	DB	85.3	85.2	86.2	87.6	89.0	89.7	90.0	91.3	90.0	88.9	87.4	86.4	
		MCWB	80.1	79.8	80.3	81.1	81.8	82.0	81.4	81.8	81.6	81.2	81.2	80.6	
	5%	DB	84.5	84.4	85.4	86.8	88.2	88.6	88.5	89.8	88.6	87.7	86.5	85.5	
		MCWB	79.7	79.4	79.9	80.8	81.5	81.7	81.1	81.6	81.2	80.9	80.7	80.3	
	10%	DB	83.5	83.5	84.5	86.1	87.3	87.5	87.2	88.2	87.3	86.6	85.7	84.4	
		MCWB	79.1	78.9	79.5	80.5	81.1	81.3	80.7	81.0	80.8	80.5	80.4	79.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.6	81.4	81.8	82.4	82.8	83.2	82.8	83.5	83.0	82.7	82.4	81.8
			MCDB	85.4	85.0	85.7	87.0	88.6	88.7	88.9	90.9	89.9	88.6	87.3	86.1
2%		WB	80.8	80.6	81.0	81.8	82.3	82.5	82.0	82.6	82.1	81.8	81.7	81.2	
		MCDB	84.6	84.3	85.1	86.4	87.7	88.3	88.1	89.6	88.4	87.3	86.5	85.3	
5%		WB	80.1	79.9	80.5	81.3	81.8	81.9	81.5	81.8	81.6	81.3	81.1	80.7	
		MCDB	83.7	83.5	84.7	86.0	87.1	87.4	87.4	88.4	87.5	86.6	85.7	84.7	
10%		WB	79.3	79.2	79.9	80.8	81.3	81.4	80.9	81.3	81.0	80.8	80.6	80.1	
		MCDB	82.7	82.7	83.8	85.4	86.5	86.8	86.5	87.3	86.6	85.8	85.1	83.8	
Mean Daily Temperature Range		5% DB	MDBR	7.6	8.3	9.1	10.1	10.7	10.7	10.9	11.8	11.3	10.7	9.4	7.9
			MCDBR	9.5	9.7	10.5	11.2	12.0	12.5	13.1	14.1	13.3	12.4	10.9	9.6
	MCWBR		5.9	6.1	6.3	6.3	6.3	6.2	6.4	6.7	6.4	6.3	6.1	5.4	
	5% WB	MCDBR	8.9	9.2	10.0	10.6	11.5	11.7	12.1	13.0	12.4	11.6	10.4	9.1	
		MCWBR	5.8	6.2	6.4	6.5	6.3	6.2	6.4	6.7	6.5	6.3	6.1	5.4	
Clear-Sky Solar Irradiance	taub	0.477	0.451	0.443	0.431	0.427	0.422	0.424	0.446	0.437	0.449	0.450	0.472		
	taud	2.242	2.333	2.378	2.418	2.455	2.480	2.471	2.391	2.414	2.382	2.368	2.269		
	Ebn at Noon	263	275	279	279	275	274	274	271	276	272	269	261		
	Edh at Noon	44	41	40	38	36	35	35	39	38	39	38	42		
All-Sky Solar Radiation	RadAvg	1187	1476	1706	1859	1807	1639	1535	1610	1554	1485	1340	1197		
	RadStd	171	187	183	134	134	101	121	100	141	106	127	102		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page